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United States Patent	12415210
Kind Code	B2
Date of Patent	September 16, 2025
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Auto-cleaning of smart air ionizer

Abstract

A cleaning mechanism for a smart air ionizer is provided. The cleaning mechanism includes an electrode configured with a plurality of carbon bristles exposed to an airflow and configured to ionize air in the airflow via the plurality of carbon bristles, a vibration mechanism, and a controller circuit coupled to the vibration mechanism, the controller circuit including a controller configured to operate the vibration mechanism to clean the plurality of carbon bristles.

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Appl. No.: 18/343654

Filed: June 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240351075 A1	Oct. 24, 2024

Foreign Application Priority Data

IN	202341029410	Apr. 24, 2023
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Publication Classification

Int. Cl.: H05F3/00 (20060101); A61L9/22 (20060101); B08B7/02 (20060101); B08B13/00 (20060101); B64D13/06 (20060101)

U.S. Cl.:

CPC B08B7/02 (20130101); A61L9/22 (20130101); B08B13/00 (20130101); B64D13/06 (20130101); A61L2209/111 (20130101); B64D2013/067 (20130101)

Field of Classification Search

USPC: None

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to, and the benefit of, India Provisional Patent Application No. 202341029410, filed Apr. 24, 2023 (DAS Code 4AB2) and titled “AUTO-CLEANING OF SMART AIR IONIZER,” which is incorporated by reference herein in its entirety for all purposes.

FIELD

(2) The present disclosure generally relates to systems for managing air quality in an aircraft, and

more specifically, to automatically cleaning air ionizers in an aircraft.

BACKGROUND

(3) Travel within aircraft includes the recirculation of air within the aircraft. Circulation of air within an enclosed space, such as aircraft, may include circulation of harmful pathogens (e.g., viruses, bacteria, etc.) and undesirable odors. Currently, commercial aircraft use air filters, including high efficiency particulate air (HEPA) filters, to filter the air during recirculation. Generally, air filters are placed at central locations and therefore may not have continuous effectiveness within the enclosed space. Furthermore, more effective air filters, such as HEPA filters, come with an increased cost both in terms of the filter itself as well as the energy used to force air through the air filter. Furthermore, air filters, including HEPA filters, do not remove odors from the air within the enclosed space.

SUMMARY

- (4) A cleaning mechanism for a smart air ionizer is disclosed herein. The cleaning mechanism includes an electrode configured with a plurality of carbon bristles exposed to an airflow and configured to ionize air in the airflow via the plurality of carbon bristles; a vibration mechanism; and a controller circuit coupled to the vibration mechanism, the controller circuit including a controller configured to operate the vibration mechanism to clean the plurality of carbon bristles.
- (5) In various embodiments, the cleaning mechanism further includes an ion probe connected to the controller. In various embodiments, the controller is configured to receive a detected ion count from the ion probe and operate the vibration mechanism based on the detected ion count.
- (6) In various embodiments, the controller operates the vibration mechanism based on a predetermined ion threshold. In various embodiments, the controller sends command to the vibration mechanism to clean the plurality of carbon bristles in response to at least one of the detected ion count failing to meet the predetermined ion threshold or the detected ion count failing to exceed the predetermined ion threshold.
- (7) In various embodiments, the vibration mechanism is coupled to the electrode. In various embodiments, the vibration mechanism cleans the plurality of carbon bristles through vibrating the electrode. In various embodiments, the vibration mechanism vibrates the electrode at a predetermined frequency. In various embodiments, the vibration mechanism is at least one of an eccentric rotating mass (ERM) motor or a printed circuit board (PCB) motor. In various embodiments, cleaning the plurality of carbon bristles ejects particulates that have accumulated on the plurality of carbon bristles.
- (8) Also disclosed herein is a passenger service unit for use above an airline seat. The passenger service unit includes a body, an air outlet mounted to the body, an electrode mounted adjacent the air outlet, the electrode configured with a plurality of carbon bristles exposed to an airflow to ionize air in the airflow, a vibration mechanism, an air ionizer circuit connected to the electrode, a processor coupled to the air ionizer circuit, and a memory operatively coupled to the processor. In various embodiments, the memory includes instructions stored thereon that, when executed by the processor, cause the processor to: receive a detected ion count from an ion probe, determine whether the detected ion count meets or exceeds a predetermined ion threshold, and, responsive to at least one of detected the ion count failing to meet the predetermined ion threshold or the detected ion count failing to exceed the predetermined ion threshold, send a command to the vibration mechanism to clean the plurality of carbon bristles.
- (9) In various embodiments, the vibration mechanism is coupled to the electrode. In various embodiments, the vibration mechanism cleans the plurality of carbon bristles through vibrating the electrode. In various embodiments, the vibration mechanism vibrates the electrode at a predetermined frequency. In various embodiments, the vibration mechanism is at least one of an eccentric rotating mass (ERM) motor or a printed circuit board (PCB) motor. In various embodiments, cleaning the plurality of carbon bristles ejects particulates that have accumulated on the plurality of carbon bristles.

(10) Additionally disclosed herein is a method for controlling an air ionizer. The method includes receiving, by a processor, a detected ion count from a probe, the detected ion count being representative of ions produced by an electrode; comparing, by the processor, the detected ion count to a predetermined ion threshold; and, responsive to the detected ion count being below the predetermined ion threshold, sending, by the processor, a command to a vibration mechanism to clean a plurality of carbon bristles coupled to the electrode.

(11) In various embodiments, the vibration mechanism cleans the plurality of carbon bristles through vibrating the electrode. In various embodiments, the vibration mechanism vibrates the electrode at a predetermined frequency. In various embodiments, the vibration mechanism is at least one of an eccentric rotating mass (ERM) motor or a printed circuit board (PCB) motor. In various embodiments, cleaning the plurality of carbon bristles ejects particulates that have accumulated on the plurality of carbon bristles.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosure, however, may best be obtained by referring to the following detailed description and claims in connection with the following drawings. While the drawings illustrate various embodiments employing the principles described herein, the drawings do not limit the scope of the claims.

(2) FIG. 1 illustrates an aircraft and various sections within the aircraft, in accordance with various embodiments.

(3) FIG. 2 illustrates a lavatory in an aircraft with a smart air ionizer, in accordance with various embodiments.

(4) FIG. 3 illustrates a cabin area in an aircraft with a smart air ionizer system, in accordance with various embodiments.

(5) FIGS. 4A and 4B illustrate a smart ionizer install in an aircraft, in accordance with various embodiments.

(6) FIG. 5 illustrates a perspective view of a smart air ionizer unit, in accordance with various embodiments.

(7) FIG. 6A illustrates an enlarged view of a carbon brush, in accordance with various embodiments.

(8) FIG. 6B illustrates an enlarged view of a plurality of carbon bristles with accumulated particulates, in accordance with various embodiments.

(9) FIG. 7 illustrates an enlarged diagram of a cleaning mechanism coupled to and in physical contact with a high voltage wire, in accordance with an illustrative embodiment.

(10) FIG. 8 illustrates a system diagram of a smart air ionizer, in accordance with various embodiments.

(11) FIG. 9 illustrates a method for controlling an auto-cleaning mechanism of a smart air ionizer, in accordance with various embodiments

DETAILED DESCRIPTION

(12) The following detailed description of various embodiments herein makes reference to the accompanying drawings, which show various embodiments by way of illustration. While these various embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, it should be understood that other embodiments may be realized and that changes may be made without departing from the scope of the disclosure. Thus, the detailed description herein is presented for purposes of illustration only and not of limitation. Furthermore, any

reference to singular includes plural embodiments and any reference to more than one component or step may include a singular embodiment or step. Also, any reference to attached, fixed, connected, or the like may include permanent, removable, temporary, partial, full or any other possible attachment option. Additionally, any reference to without contact (or similar phrases) may also include reduced contact or minimal contact. It should also be understood that unless specifically stated otherwise, references to “a,” “an” or “the” may include one or more than one and that reference to an item in the singular may also include the item in the plural. Further, all ranges may include upper and lower values and all ranges and ratio limits disclosed herein may be combined.

(13) Disclosed herein are mechanisms for auto-cleaning a smart air ionizer. In various embodiments, the smart air ionizer may have a dual capability of neutralizing pathogens (e.g., bacteria, viruses, molds, etc.) and neutralizing undesirable odors (e.g., volatile organics, sulfur-based compounds, etc.). In various embodiments, the smart air ionizer may be used alongside existing air filters. In various embodiments, a high efficiency particulate air (HEPA) filter for improved performance. In various embodiments, the combination of the smart air ionizer and the air filter may perform at a higher minimum efficiency reporting value (MERV) rating than the MERV rating of the air filter. Accordingly, a non-HEPA filter paired with the smart air ionizer may perform with a higher MERV value than a HEPA filter. In various embodiments, the smart air ionizer ionizes the surrounding air, generating positive and negative ions. The positive and negative ions combine with pathogens and odor molecules in the air and effectively neutralizes both.

(14) In various embodiments, the ionization of the air is based on ion emitters, i.e. carbon brushes. In various embodiments, the carbon brushes are configured with a plurality of carbon bristles, with a structure of each carbon bristle causing the surrounding air to conduct electricity, i.e. a dielectric breakdown. In that regard, in various embodiments, a topology of the plurality of carbon bristles is critical to the generation of ions. In various embodiments, the plurality of carbon bristles, when in operation, are exposed to surrounding environment and attract particulates. In various embodiments, the sedimentation of such particulates on and/or in the plurality of carbon bristles affect the efficiency of ion emission of the plurality of carbon bristles. Typically, ions created by air ionizer, such as negative ions and positive ions, have an active life span of about 30 seconds to about 75 seconds. In that regard, in various embodiments, there is a need for efficient distribution in air space and, by increasing the reach of these ions, a large volume of air may be covered.

(15) Accordingly, in various embodiments, mechanisms are provided for auto-cleaning a smart air ionizer. In various embodiments, a cleaning mechanism is coupled to each support electrode, i.e. each electrode assembly that supports its associated plurality of carbon brushes. In various embodiments, in operation, each cleaning mechanism is in physical contact with electrode assembly and vibrates at a predetermined frequency. In that regard, in various embodiments, each electrode assembly is free to move, i.e. are not constrained. Accordingly, in various embodiments, particulates that accumulate on the plurality of carbon bristles associated with the electrode assembly are ejected from the plurality of carbon bristles due to one or more of a vibratory force, an inertial force, and/or a fictional force.

(16) Referring now to FIG. 1, an aircraft **100** and various sections within the aircraft is illustrated, in accordance with various embodiments. Aircraft **100** is an example of a passenger or transport vehicle in which smart air ionizers may be implemented in accordance with various embodiments. In various embodiments, aircraft **100** has a starboard wing **102** and a port wing **104** attached to a fuselage **106**. In various embodiments, aircraft **100** also includes a starboard engine **108** connected to starboard wing **102** and a port engine **110** connected to port wing **104**. In various embodiments, aircraft **100** also includes a starboard horizontal stabilizer **112**, a port horizontal stabilizer **114**, and a vertical stabilizer **116**. In various embodiments, aircraft **100** also includes various cabin sections, including, for example, a first cabin section **118**, a second cabin section **120**, a third cabin section **122**, and a pilot cabin **124**. In various embodiments, aircraft **100** may include a front lavatory **126**

and/or a rear lavatory **128**.

(17) Referring now to FIG. **2**, a lavatory **200** is illustrated, in accordance with various embodiments. In various embodiments, lavatory **200** may be an example of front lavatory **126** or rear lavatory **128**. Lavatory **200** may include a sink **202**, a toilet **204**, a trash receptacle **206**, an air vent **208**, and an air ionizer **210**. In various embodiments, air ionizer **210** may be mounted on ceiling **216** of lavatory **200**, adjacent to air vent **208**. In various embodiments, air ionizer **210** may be mounted within air vent **208**. Air ionizer **210** generates ionized air particles, including negative ions **212** and positive ions **214**, that are spread throughout lavatory **200** by air flow from air vent **208**.

(18) In various embodiments, air ionizer **210**, and more specifically, ions created by air ionizer **210** may neutralize pathogens (e.g., bacteria, viruses, molds, dust, etc.) and/or malodor (i.e., unpleasant smells) that are airborne and on surfaces. Pathogens and malodor may be generated and/or spread by sink **202**, toilet **204**, and trash receptacle **206**, among other locations. Pathogens and malodor may be airborne and/or settle on surfaces within lavatory **200**, including sink **202**, toilet **204**, and trash receptacle **206**, among others.

(19) As illustrated in FIG. **2**, air ionizer **210** may be placed adjacent to air vent **208** so that air ionizer **210** ionizes the air exiting the air vent **208**. Generally, ions created by air ionizer **210** have an active life span of about 30 seconds to about 75 seconds, and more specifically, about 45 seconds to about 60 seconds. Accordingly, locating the air ionizer **210** adjacent to air vent **208** improves the efficacy of ions as compared to placing an air ionizer within ductwork of an air handling system or adjacent an air filter within the air handling system. In various embodiments, air ionizer **210** may be placed inside air vent **208**, and more specifically, within the ductwork behind air vent **208** and immediately adjacent to air vent **208**. In various embodiments, air ionizer **210** may be incorporated into air vent **208** so the air ionizer **210** and air vent **208** operate as a single unit to ionize and disperse air throughout lavatory **200**.

(20) Referring now to FIG. **3**, a cabin **300** of an aircraft (e.g., aircraft **100**) is illustrated, in accordance with various embodiments. Cabin **300** may be an example of first cabin section **118**, second cabin section **120**, and/or third cabin section **122**. Cabin **300** includes a floor **302**, one or more rows of passenger seats **304** mounted to floor **302**, a fuselage **306** surrounding the cabin **300** to form outer walls and ceiling of cabin **300**, and overhead bins **316** that are mounted to fuselage **306**. Cabin **300** further includes passenger air outlets **308** are mounted underneath overhead bins **316** and above one or more of the passenger seats **304**. In various embodiments, passenger air outlet **308** may be an air grasper that is commonly placed above the passenger seats **304** and is adjustable by rotating a nozzle on the air grasper. In various embodiments, there may be one passenger air outlet **308** for each passenger seat **304**. Cabin **300** further includes cabin air outlets **318** that conduct, or provide, air from above overhead bins **316**. Air in cabin **300** is circulated (as indicated by the arrows) from passenger air outlets **308** and cabin air outlets **318** throughout cabin **300**, including over and around, passenger seats **304**, and into one or more air returns **320**. In various embodiments, air may circulate through air ducts **322** and through air filters before returning into cabin **300** through passenger air outlets **308** and cabin air outlets **318**.

(21) As illustrated in FIG. **3**, air ionizer **310** is placed adjacent to the passenger air outlet **308** to ionize the air as it exits the passenger air outlet **308**. In various embodiments, there may be one air ionizer **310** for each passenger air outlet **308**. In various embodiments, there may be one air ionizer **310** for each row of the passenger seats **304**, where each passenger seat **304** is associated with one passenger air outlet **308**. In various embodiments, one or more air ionizers **310** may be placed adjacent to cabin air outlets **318**. In various embodiments, air ionizer **310** may be placed inside air system ductwork adjacent to the passenger air outlet **308** and cabin air outlet **318**. In various embodiments, air ionizer **310** may be incorporated into a passenger service unit (PSU) within cabin **300**.

(22) Referring now to FIGS. **4A** and **4B**, illustrated are a passenger service unit (PSU) **400** and

PSU **450** that may be used in an aircraft cabin (e.g., cabin **300**), in accordance with various embodiments. FIG. **4A** illustrates PSU **400** above two passenger seats **404**, PSU **400** including two passenger air outlets **408**, an air ionizer **410**, and two reading lights **430**. As illustrated, air ionizer **410** is located on PSU **400** and adjacent to the two passenger air outlets **408**. Placing the air ionizer **410** adjacent to the two passenger air outlets **408** improves the performance of air ionizer **410** by increasing the amount of time the ions created by air ionizer **410** are circulated through cabin **300**. In various embodiments, air ionizer **410** may be located inside PSU **400** and adjacent to the two passenger air outlets **408**. Placing the air ionizer **410** inside PSU **400** may be more visually appealing with little to no degradation in the performance of air ionizer **410**. In various embodiments, PSU **400** may be installed above a two-seat aisle in an aircraft.

(23) FIG. **4B** illustrates PSU **450** including a body **452** that includes three passenger air outlets **458**, an air ionizer **460**, three passenger reading lights **462**, and three call buttons **464**. Air ionizer **460** is placed on an outside of the body **452** adjacent the three passenger air outlets **458** to provide improved ionizing performance. In various embodiments, air ionizer **460** may be placed on an inside of the body **452** and adjacent to the passenger air outlets **458** with little to no degradation in performance of air ionizer **410**. In various embodiments, PSU **450** may be installed above a three-seat aisle in an aircraft.

(24) Referring now to FIG. **5**, illustrated is a smart air ionizer **510**, in accordance with various embodiments. Smart air ionizer **510** may be an example of air ionizer **210** described above with respect to FIG. **2**, air ionizer **310** described above with respect to FIG. **3**, air ionizer **410** described above with respect to FIG. **4A**, or air ionizer **460** described above with respect to FIG. **4B**. In various embodiments, smart air ionizer **510** includes an enclosure **512**, an air ionizer circuit **514**, a high voltage circuit **516**, a mounting flange **518**, and an input power cable **520**. In various embodiments, high voltage circuit **516** is connected to air ionizer circuit **514** and both are disposed within enclosure **512**. In various embodiments, enclosure **512** may be mounted to a ceiling (e.g., ceiling **216**), a wall, a passenger service unit (e.g., PSU **400**, PSU **450**), or an air duct, among other locations. In various embodiments, mounting flange **518** may be used to secure enclosure **512** to the surface on which it is mounted.

(25) In various embodiments, smart air ionizer **510** further includes a first high voltage wire **522a**, a second high voltage wire **522b**, a first carbon brush **524a**, and a second carbon brush **524b**. In various embodiments, first high voltage wire **522a** is connected to air ionizer circuit **514** at a first end and to first carbon brush **524a** at a second end. In various embodiments, first high voltage wire **522a** and first carbon brush **524a** may be collectively referred to as a first electrode **526a**. In various embodiments, second high voltage wire **522b** is connected to air ionizer circuit **514** at a first end and to second carbon brush **524b** at a second end. In various embodiments, second high voltage wire **522b** and second carbon brush **524b** may be collectively referred to as a second electrode **526b**. Also illustrated are an air flow **F1** into smart air ionizer **510** and an ionized air flow **F2** out smart air ionizer **510**.

(26) Input power cable **520** provides power for smart air ionizer **510**. In various embodiments, input power cable **520** may provide electrical power, for example, 115 VAC. In various embodiments, input power cable **520** may provide 28 VDC. In various embodiments, another AC or DC voltage may be provided. Power provided by input power cable **520** may be used to power the air ionizer circuit **514** and high voltage circuit **516**.

(27) Air ionizer circuit **514** includes a controller that is configured to control the various features and elements of the smart air ionizer. Air ionizer circuit **514** may be configured for use with various external high voltage ranges and electrode types (e.g., first and second high voltage wires **522a**, **522b** and first and second carbon brushes **524a**, **524b**). The controller may include one or more processors configured to implement various logical operations in response to execution of instructions, for example, instructions stored on a non-transitory, tangible, computer-readable medium. The one or more processors can be a general-purpose processor, a microprocessor, a

microcontroller, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete or transistor logic, discrete hardware components, or any combination thereof. The controller may further include memory to store data, executable instructions, system program instructions, and/or controller instructions to implement the control logic of the controller.

(28) High voltage circuit **516** may include a switching regulator, a transformer, and a passive voltage tripler. In various embodiments, high voltage circuit **516** may receive input voltage from input power cable **520**. In various embodiments, high voltage circuit **516** may output about 2.5 KV to about 12.5 KV, and more specifically, about 5 KV to about 10 KV. In various embodiments, the controller in air ionizer circuit **514** may control the high voltage circuit **516**. In various embodiments, the controller in air ionizer circuit **514** may control the output voltage of high voltage circuit **516**. In various embodiments, an output current of high voltage circuit **516** may be limited to microamps.

(29) The high voltage output of high voltage circuit **516** is sent to first electrode **526a** and second electrode **526b**. That is, the high voltage is sent to first high voltage wire **522a** and second high voltage wire **522b** and subsequently to first carbon brush **524a** and second carbon brush **524b**, respectively. Together, first electrode **526a** and second electrode **526b** use the high voltage output of high voltage circuit **516** to ionize the air surrounding first carbon brush **524a** and second carbon brush **524b**, respectively. Air ionizer circuit **514**, and more specifically, the controller of air ionizer circuit **514** controls the voltage and current sent to first and second electrodes **526a**, **526b**.

(30) With temporary reference to FIG. **6A**, an enlarged view of a carbon brush, such as a first carbon brush **524a**, and a second carbon brush **524b** of FIG. **5**, is illustrated in accordance with various embodiments. In various embodiments, carbon brush **624**, which is coupled to high voltage wire **622**, is a plurality of carbon bristles **628**, with a structure of each carbon bristle causing the surrounding air to conduct electricity, i.e. a dielectric breakdown. In various embodiments, a topology of the plurality of carbon bristles **628** topology is important to the generation of ions. Accordingly, in various embodiments, the plurality of carbon bristles, when in operation, are exposed to surrounding environment and attract particulates. With temporary reference to FIG. **6B**, an enlarged view of a plurality of carbon bristles with accumulated particulates is illustrated, in accordance with various embodiments. In various embodiments, when high voltage wire **622** is receiving power and the plurality of carbon bristles **628** are exposed to surrounding environment, particulates **630** accumulate on and/or in the plurality of carbon bristles, which affects the efficiency of ion emission of the plurality of carbon bristles **628**.

(31) In order to automatically clean the plurality of carbon bristles **628**, in various embodiments, the controller in air ionizer circuit **514** may be configured to further control the first cleaning mechanism **532a** and second cleaning mechanism **532b**. In various embodiments, the first cleaning mechanism **532a** and the second cleaning mechanism **532b** may be a vibration mechanism. In various embodiments, the first cleaning mechanism **532a** and the second cleaning mechanism **532b** may be an eccentric rotating mass (ERM) motor or a printed circuit board (PCB) motor, among others. In various embodiments, the first cleaning mechanism **532a** is coupled to and in physical contact with the first high voltage wire **522a** and the second cleaning mechanism **532b** is coupled to and in physical contact with second high voltage wire **522b**. In various embodiments, the first cleaning mechanism **532a** and the second cleaning mechanism **532b**, under the control of the controller of the air ionizer circuit **514**, are configured to vibrate at a predetermined frequency. In various embodiments, the controller of the air ionizer circuit **514** may be configured to turn the first cleaning mechanism **532a** and the second cleaning mechanism **532b** on or off based in input from an ion probe that detecting the number of ions produced by first and second electrodes **526a**, **526b**. In that regard, in various embodiments, responsive to the first cleaning mechanism **532a** and the second cleaning mechanism **532b**, along with their respective high voltage wires, i.e. the first high voltage wire **522a** and second high voltage wire **522b**, being activated by the controller of the air

ionizer circuit **514**, the first cleaning mechanism **532a** and the second cleaning mechanism **532b**, along with their respective high voltage wires, i.e. the first high voltage wire **522a** and second high voltage wire **522b**, are free to move independently, i.e. are not constrained, so that any particulates, i.e. particulates **630** of FIG. **6B**, that accumulate on the plurality of carbon bristles **628** are ejected from the plurality of carbon bristles **628** due to one or more of a vibratory force, an inertial force, and/or a fictional force. In various embodiments, the ejected particulates may then be captured in an adjacent air filter within the air handling system with which the smart air ionizer **510** is associated.

(32) Referring now to FIG. **7**, an enlarged diagram of a cleaning mechanism coupled to and in physical contact with a high voltage wire is illustrated, in accordance with illustrative embodiments. In various embodiments, the cleaning mechanism **532**, such as the first cleaning mechanism **532a** and the second cleaning mechanism **532b** of FIG. **5**, is coupled to and in physical contact with high voltage wire **522**, such as the first high voltage wire **522a** or the second high voltage wire **522b** of FIG. **5**, via coupler **702**. In various embodiments, the cleaning mechanism **532** is a motor **704** with an offset (non-symmetric) mass **706** attached to a shaft **708** of the motor **704**. In various embodiments, as the motor **704** rotates, the centripetal force of the offset mass **706** is asymmetric, resulting in a net centrifugal force, and this causes a displacement of the motor **704** and thus, the high voltage wire **522**. In various embodiments, the displacement of the high voltage wire **522** via the displacement of the motor **704** causes one or more of a vibratory force, an inertial force, and/or a fictional force to be placed on the plurality of carbon bristles **628** and thereby an ejection of any accumulated particulates on the plurality of carbon bristles **628**.

(33) Referring now to FIG. **8**, a diagram of system **800** of an air ionizer circuit is illustrated, in accordance with various embodiments. System **800** may be an example of air ionizer circuit **514** and high voltage circuit **516** described above with respect to FIG. **5**. System **800** includes a power input **802**, a protection module **804**, an active clamp **806**, and a converter **808**, which may be collectively referred to as a power circuit. In various embodiments, power input **802** may provide 18 VDC to 32 VDC, and more specifically, may provide 28 VDC to system **800**. In various embodiments, power input **802** may provide 110 VAC to 240 VAC, and more specifically, may provide 115 VAC to system **800**. In various embodiments, system **800** may optionally include an AC-DC converter to convert AC voltage input to DC voltage. In various embodiments, protection module **804** may provide electrostatic discharge (ESD) and/or surge protection for system **800**. In various embodiments, active clamp **806** may provide protection for the power supply (e.g., power input **802**) against back electric and magnetic fields (EMF). In various embodiments, converter **808** may be a DC-DC voltage converter. In various embodiments, converter **808** may be a buck-boost converter to step-down DC voltage or step-up DC voltage.

(34) System **800** further includes a transformer **810**, a high voltage controller **812**, a regulator **814**, a multiplier **816**, a current limiter **818**, a feedback circuit **820**, a controller **822**, a feedback switch **834**, a first electrode **826a**, a second electrode **826b**, first cleaning mechanism **832a**, and the second cleaning mechanism **832b**. In various embodiments, transformer **810** may be an isolation transformer driver that provides separation from power input **802** and high voltage controller **812** first and second electrodes **826a**, **826b**. In various embodiments, transformer **810** may receive a voltage from converter **808** to be provided to multiplier **816**. In various embodiments, multiplier **816** may multiply, or step-up, the voltage provided by transformer **810** to a high voltage for use by first and second electrode **826a**, **826b**. Multiplier **816** provides the high voltage to current limiter **818** that supplies the high voltage and limited current to first and second electrode **826a**, **826b** while maintaining the high voltage.

(35) In various embodiments, the high voltage provided to first and second electrodes **826a**, **826b** may be about 2.5 KV to about 12.5 KV, and more specifically, about 5 KV to about 10 KV. In various embodiments, the current provided to first and second electrodes **826a**, **826b** may be about 100 μ A to about 1 mA, and more specifically, about 250 μ A to about 750 μ A. By applying a high

voltage with a limited current to first and second electrode **826a**, **826b**, little to no ozone is produced by first and second electrodes **826a**, **826b** during the air ionizing process. Producing little to no ozone provides increased safety for use in enclosed spaces (e.g., aircraft **100**) and is an improvement over existing systems.

(36) There is a first feedback path between current limiter **818** and feedback circuit **820** that provides the voltage and current output of current limiter **818** is provided to feedback circuit **820**. In various embodiments, there may be a second feedback path between transformer **810** and feedback circuit **820** that provides the voltage and current output of transformer **810** to feedback circuit **820**. Feedback circuit **820** provides the voltage and current output to high voltage controller **812** and controller **822**. The feedback provided allows high voltage controller **812** and controller **822** to vary output voltage, output current, and ion count and provide improved performance of system **800**.

(37) High voltage controller **812** receives power from converter **808** and is configured to control the transformer **810** based on determined operation parameters (e.g., ion count) and feedback received from feedback circuit **820**. In various embodiments, regulator **814** may receive power from converter **808** and provide a power supply to controller **822**. Controller **822** may communicate with high voltage controller **812** to provide the determined operation parameters (e.g., ion count). In various embodiments, controller **822** may be configured to receive input and provide output for system **800**. In various embodiments, controller **822** may send instructions to high voltage controller **812** to change operating parameters.

(38) In various embodiments, controller **822** may be configured to, based on the determined operation parameters (e.g., ion count), turn the first cleaning mechanism **832a** and the second cleaning mechanism **832b** on or off in order to improve the ion count. In that regard, responsive to the ion count falling below a predetermined level, the controller **822** may turn on the first cleaning mechanism **832a** and the second cleaning mechanism **832b** in order to eject any particulates that may have accumulated on the plurality of carbon bristles through one or more of a vibratory force, an inertial force, and/or a fictional force.

(39) System **800** further includes an input **836**, an output **838**, a first communication bus **840**, and a second communication bus **842**. In various embodiments, input **836** may be a keypad. In various embodiments, input **836** may be a button or a switch. In various embodiments, input **836** may be a touch screen interface. In various embodiments, input **836** may further include an ion probe for detecting the number of ions produced by first and second electrodes **826a**, **826b**. In various embodiments, output **838** may be a screen, an LED, or light, among others. In various embodiments, input **836** and output **838** may be combined in a touch screen. In various embodiments, first communication bus **840** and second communication bus **842** may be a serial peripheral interface (SPI) bus, a universal asynchronous receiver-transmitter (UART) bus, a controller area network (CAN) bus, or an ethernet connection, among others.

(40) Controller **822** may include one or more processors configured to implement various logical operations in response to execution of instructions, for example, instructions stored on a non-transitory, tangible, computer-readable medium. The one or more processors can be a general-purpose processor, a microprocessor, a microcontroller, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete or transistor logic, discrete hardware components, or any combination thereof. Controller **822** may further include memory to store data, executable instructions, system program instructions, and/or controller instructions to implement the control logic of controller **822**.

(41) High voltage controller **812** may include one or more controllers configured to implement various logical operations in response to execution of instructions, for example, instructions stored on a non-transitory, tangible, computer-readable medium. The one or more controllers can be a voltage switching regulator, a pulse width modulation (PWM) controller, a PWM driver, an

application specific integrated circuit (ASIC), discrete or transistor logic, discrete hardware components, or any combination thereof, among others. High voltage controller **812** may further include memory to store data, executable instructions, system program instructions, and/or controller instructions to implement the control logic of high voltage controller **812**.

(42) Referring now to FIG. **9**, a method **900** for controlling an auto-cleaning mechanism of a smart air ionizer is illustrated, in accordance with various embodiments. In various embodiments, the smart air ionizer may be smart air ionizer **510** or system **800** described above with respect to FIGS. **5** and **8**, respectively. In various embodiments, method **900** may be performed by air ionizer circuit **514** or controller **822** described above with respect to FIGS. **5** and **8**.

(43) At decision block **902**, controller **822** determines whether an ion count has been received. In various embodiments, the ion count may be received from an ion probe connected to controller **822**. In various embodiments, the ion count may be measured as a total number of ions or as a sample number of ions.

(44) If at decision block **902** it is determined that an ion count has been received, then, at decision block **904**, controller **822** determines whether the number of ions counted is less than an expected number of ions, an ion threshold. If at decision block **904** it is determined that the number of ions is less than the number of ions expected, at block **906**, controller **822** sends a command to turn on the vibration mechanisms for a set time period, such as the first cleaning mechanism **532a** and the second cleaning mechanism **532b** of FIG. **5** or the first cleaning mechanism **832a** and the second cleaning mechanism **832b** of FIG. **8**.

(45) At decision block **908**, controller **822** determines whether the set time period has expired. If at decision block **908** the set time period has not expired, the method **900** returns to decision block **908**. If at decision block **908** the set time period has expired, controller **822** sends a command to turn off the vibration mechanisms.

(46) Returning to decision block **904**, if it is instead determined that the number of ions counted meets or exceeds the number of ions expected, method **900** returns to decision block **902**. Method **900** then proceeds as described above.

(47) Returning to decision block **902**, if it is instead determined that an ion count has not been received, then, at decision block **912**, controller **822** determines whether a predetermined time period has expired. If at decision block **912** the predetermined time period has not expired, the method **900** returns to decision block **902**. If at decision block **912** the predetermined time period has expired, the method proceeds to decision block **908**.

(48) System program instructions and/or controller instructions may be loaded onto a non-transitory, tangible computer-readable medium having instructions stored thereon that, in response to execution by a controller, cause the controller to perform various operations. The term “non-transitory” is to be understood to remove only propagating transitory signals per se from the claim scope and does not relinquish rights to all standard computer-readable media that are not only propagating transitory signals per se.

(49) Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may

be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. Different cross-hatching is used throughout the figures to denote different parts but not necessarily to denote the same or different materials.

(50) Systems, methods and apparatus are provided herein. In the detailed description herein, references to “one embodiment,” “an embodiment,” “various embodiments,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic.

Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

(51) Numbers, percentages, or other values stated herein are intended to include that value, and also other values that are about or approximately equal to the stated value, as would be appreciated by one of ordinary skill in the art encompassed by various embodiments of the present disclosure. A stated value should therefore be interpreted broadly enough to encompass values that are at least close enough to the stated value to perform a desired function or achieve a desired result. The stated values include at least the variation to be expected in a suitable industrial process, and may include values that are within 5% of a stated value. Additionally, the terms “substantially,” “about” or “approximately” as used herein represent an amount close to the stated amount that still performs a desired function or achieves a desired result. For example, the term “substantially,” “about” or “approximately” may refer to an amount that is within 5% of a stated amount or value.

(52) Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112 (f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

(53) Finally, it should be understood that any of the above-described concepts can be used alone or in combination with any or all of the other above-described concepts. Although various embodiments have been disclosed and described, one of ordinary skill in this art would recognize that certain modifications would come within the scope of this disclosure. Accordingly, the description is not intended to be exhaustive or to limit the principles described or illustrated herein to any precise form. Many modifications and variations are possible in light of the above teaching.

Claims

1. A cleaning mechanism for a smart air ionizer, comprising: an electrode configured with a plurality of carbon bristles exposed to an airflow and configured to ionize air in the airflow via the plurality of carbon bristles; a vibration mechanism; a controller circuit coupled to the vibration mechanism, the controller circuit including a controller configured to operate the vibration mechanism to clean the plurality of carbon bristles; and an ion probe connected to the controller, wherein the controller is configured to receive a detected ion count from the ion probe and operate the vibration mechanism based on the detected ion count.

2. The smart air ionizer of claim 1, wherein the controller operates the vibration mechanism based on a predetermined ion threshold.

3. The smart air ionizer of claim 2, wherein the controller sends command to the vibration mechanism to clean the plurality of carbon bristles in response to at least one of the detected ion count failing to meet the predetermined ion threshold or the detected ion count failing to exceed the predetermined ion threshold.
 4. The smart air ionizer of claim 1, wherein the vibration mechanism is coupled to the electrode.
 5. The smart air ionizer of claim 1, wherein the vibration mechanism cleans the plurality of carbon bristles through vibrating the electrode.
 6. The smart air ionizer of claim 1, wherein the vibration mechanism vibrates the electrode at a predetermined frequency.
 7. The smart air ionizer of claim 1, wherein the vibration mechanism is at least one of an eccentric rotating mass (ERM) motor or a printed circuit board (PCB) motor.
 8. The smart air ionizer of claim 1, wherein cleaning the plurality of carbon bristles ejects particulates that have accumulated on the plurality of carbon bristles.
 9. A passenger service unit for use above an airline seat, comprising: a body; an air outlet mounted to the body; an electrode mounted adjacent the air outlet, the electrode configured with a plurality of carbon bristles exposed to an airflow to ionize air in the airflow; a vibration mechanism; an air ionizer circuit connected to the electrode; a processor coupled to the air ionizer circuit; and a memory operatively coupled to the processor, the memory comprising instructions stored thereon that, when executed by the processor, cause the processor to: receive a detected ion count from an ion probe; determine whether the detected ion count meets or exceeds a predetermined ion threshold; and responsive to at least one of the detected ion count failing to meet the predetermined ion threshold or the detected ion count failing to exceed the predetermined ion threshold, send a command to the vibration mechanism to clean the plurality of carbon bristles.
 10. The passenger service unit of claim 9, wherein the vibration mechanism is coupled to the electrode.
 11. The passenger service unit of claim 9, wherein the vibration mechanism cleans the plurality of carbon bristles through vibrating the electrode.
 12. The passenger service unit of claim 9, wherein the vibration mechanism vibrates the electrode at a predetermined frequency.
 13. The passenger service unit of claim 9, wherein the vibration mechanism is at least one of an eccentric rotating mass (ERM) motor or a printed circuit board (PCB) motor.
 14. The passenger service unit of claim 9, wherein cleaning the plurality of carbon bristles ejects particulates that have accumulated on the plurality of carbon bristles.
 15. A method for controlling an air ionizer, comprising: receiving, by a processor, a detected ion count from a probe, the detected ion count being representative of ions produced by an electrode; comparing, by the processor, the detected ion count to a predetermined ion threshold; and responsive to the detected ion count being below the predetermined ion threshold, sending, by the processor, a command to a vibration mechanism to clean a plurality of carbon bristles coupled to the electrode.
 16. The method of claim 15, wherein the vibration mechanism cleans the plurality of carbon bristles through vibrating the electrode.
 17. The method of claim 15, wherein the vibration mechanism vibrates the electrode at a predetermined frequency.
 18. The method of claim 15, wherein the vibration mechanism is at least one of an eccentric rotating mass (ERM) motor or a printed circuit board (PCB) motor.
 19. The method of claim 15, wherein cleaning the plurality of carbon bristles ejects particulates that have accumulated on the plurality of carbon bristles.
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